

Online Temporal Action Detection for Distress / SOS Hand Signals

M.Tech thesis — progress report

Date	2026-09-28
Repository	github.com/asthaberi/sos-oad
Commit	890cfe7 (main, uncommitted changes present)
Phase	Phase 1 — IPN Hand as engineering vehicle
Status	Stages 0–2 complete; Stage 3 in progress

1. Summary

The system under development labels the *current* frame of a continuous video stream and decides when to raise an alarm. This is online temporal action detection: there are no clip boundaries at inference time, and no module may read a frame from the future. Phase 1 uses the IPN Hand dataset as an engineering vehicle; its thirteen classes are touchless interface commands rather than distress signals, but it has the right shape — continuous video, a dominant non-gesture class, per-subject metadata and fuzzy gesture onsets. Phase 2 substitutes our own recorded SOS data behind a dataset adapter, with no change to training or evaluation code.

Work to date has been data integrity, splitting methodology and causality infrastructure. There are no experimental results yet, and none should be expected before Stage 4. This ordering is deliberate. On a subject-disjoint online detection task the characteristic failure is not a poor score but a high and meaningless one, produced by leakage that no downstream check can see. Each item below exists to make a later result defensible rather than to produce one sooner.

2. Stage plan and status

#	Content	Gate	Status
0	Repository, canonical data format, config system	Validator passes on a synthetic fixture	Complete
1	Acquire IPN Hand, write dataset adapter	Published figures reproduced	Complete
2	Frozen subject-disjoint splits and LOSGO folds	No subject in more than one split	Complete
3	Feature extraction, two streams, cached	All 200 videos cached and verified	In progress
4	Baselines B0 and B1, full evaluation harness	Both baselines produce the complete metric set	Not started
5	M1 model (causal GRU, two-stream)	—	Not started
6	Online decision layer	—	Not started
7	Metrics and headline figure	—	Not started
8	Analysis and ablations	—	Not started

3. Methodological constraints

Five constraints are enforced by automated tests rather than by convention. They are recorded here because each one, if violated, would invalidate the result rather than merely degrade it.

- **Causality.** No module may read a future frame. Enforced by feeding a truncated stream of length t and requiring the outputs for frames $0..t$ to be *bit-identical* to the corresponding slice of the full-stream run. Bit-identical rather than approximately equal: a module that is almost causal reads the future a little, and a little is enough to bias an onset-latency measurement.
- **Subject-disjoint splits.** Never split by frame, window or clip. Splits are written to disk, frozen, and placed under version control.
- **The test set is never tuned against.** Model selection uses validation only.
- **The non-gesture class is never dropped.** Removing it would silently convert online detection into trimmed-clip classification and render the numbers meaningless.
- **Provenance.** Every run records its config, seed, git commit and metrics. A result without a run directory behind it is not treated as a result.

A working tripwire accompanies these: any metric exceeding 97% is treated as a symptom of leakage and investigated before it is reported.

4. Completed work

4.1 Stage 0 — canonical data format

A single data format was specified before any code was written. Every dataset is converted into it by a small adapter, and training, evaluation and model code read nothing else. The format is enforced by a schema validator whose findings carry machine-readable codes, so tests assert on specific failures rather than on the fact that something went wrong. The central invariant is that annotations must tile each video exactly: non-gesture spans are explicit rows, never gaps inferred from absence, because inferring them is how the non-gesture class quietly disappears.

4.2 Stage 1 — dataset acquisition and verification

All published figures for IPN Hand were reproduced from the files on disk and printed beside the paper's values as the stage gate.

Quantity	Published	Computed	Agreement
Videos	200	200	exact
Subjects	50	50	exact
Gesture classes	13	13	exact
Gesture instances	4,218	4,218	exact
Non-gesture instances	1,431	1,431	exact
Frames	~800,000	800,491	within tolerance

Table II of the paper was additionally reproduced row by row — per-class instance count, mean span duration and standard deviation — for all fourteen classes. This second check matters because it verifies the class *mapping* and not merely the class *count*; see section 5.3.

Per-class composition of the converted dataset:

Class	Instances	Frames	% of frames
none	1,431	210,542	26.3
pointing with one finger	1,010	221,193	27.6
pointing with two fingers	1,007	225,683	28.2
click with one finger	200	11,197	1.4
click with two fingers	200	11,935	1.5
throw up	200	12,400	1.5
throw down	201	13,095	1.6
throw left	200	13,226	1.7
throw right	200	12,761	1.6
open twice	200	15,126	1.9
double click with one finger	200	13,525	1.7
double click with two fingers	200	13,910	1.7
zoom in	200	13,033	1.6
zoom out	200	12,865	1.6

4.3 Stage 2 — frozen subject-disjoint splits

Split	Subjects	Videos
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Train	30	120
Validation	7	28
Test	13	52

Cross-validation uses 5 leave-one-subject-group-out folds of 7/8/8/7/7 subjects over the non-test subjects (seed 1337).

Two choices here are worth the supervisor's attention. First, **the test set is the dataset authors' own partition**, not one we selected. Their partition is subject-disjoint under our subject identity, and adopting it forecloses any question about whether several splits were tried until the numbers looked favourable. Not having chosen the test set is the strongest available form of 'never tuned against'. Second, **the primary validation set is cross-validation fold 0 promoted**, rather than a separate draw. Had it been drawn separately, those subjects would appear as *training* subjects in four of the five folds, and the variance estimate reported in Stage 8 would be bounding a different experiment from the headline number it is meant to qualify.

The frozen split is committed to version control, carries a digest of the subject list it was built from, and is refused at load time if that digest no longer matches the dataset. Regenerating it requires an explicit override, which reports which subjects would change side.

5. Data-integrity findings

Three defects were found in the dataset or in the obvious way of reading it. Each would have produced a plausible-looking result rather than an error.

5.1 Subject identity is not the subject number

IPN Hand video names decompose as camera, subject, hand and clip number. The subject token is *not* an identity: it restarts at 1 for each camera and repeats across cameras, giving only 29 distinct values for 50 subjects. The pair (camera, subject) yields exactly 50 groups of exactly 4 videos, matching the published figures, and the authors' own train/test partition is disjoint under it while eight bare subject tokens straddle it. Splitting on the token alone would have placed the same person on both sides of the split, producing an inflated accuracy with no detectable symptom.

5.2 The dataset's frame counts are wrong for fourteen videos

The supplied `metadata.csv` reports 800,505 frames in total. Counting the image files gives 800,491. The discrepancy is exactly one frame for exactly fourteen videos, and those fourteen are precisely the ones whose frame directory contains a stray `Windows desktop.ini` file: the column was evidently produced by counting directory entries rather than images. Video length is therefore taken from the annotations, which match the image count for all 200 videos. Trusting the metadata would have appended one non-existent frame to each of those videos; it would have validated cleanly and produced fourteen feature vectors with no image behind them.

5.3 The class mapping was verified rather than assumed

The dataset ships class *codes* only — B0A, G01 and so on — with no statement of which code corresponds to which gesture. The names were matched positionally against the paper's class table. Because a wrong mapping would leave every total reconciling exactly, a test simulates all 91 possible pairwise label swaps and confirms that 90 of them would be detected by the per-class statistics. The single undetectable pair, which the paper rounds to identical duration figures, was resolved by rendering frames from instances of both classes and inspecting them directly.

5.4 A defect found in our own code by the causality tests

On its first run the causality suite failed twelve tests against our own extractor. The cause was not a future read: floating-point matrix multiplication is not invariant to batch size, so the same input produces slightly different output depending on how many rows share the call. On a truncated stream the final batch is smaller, and bit-identity failed for a reason unrelated to causality. The tempting response is to relax the requirement to a numerical tolerance — which would have left the project with a causality test incapable of detecting a causality violation. Instead the behaviour was characterised (only the row count matters, not position within the batch) and the final batch is now padded to a fixed size. The batch size is recorded alongside the cache because it changes the stored values.

6. Verification

Suite	Result
Total	203 passed in 54.58s
Causality	41 passed, 162 deselected in 0.58s
Splits	36 passed, 167 deselected in 7.66s

The causality suite is the load-bearing one. Besides testing the real implementation, it constructs three deliberately non-causal extractors — one using a clip centred on its own index, one anchoring the snippet grid to the final frame, one interpolating forward between snippets — and asserts that the harness *rejects*

each. All three produce output of the correct shape, correct dtype and finite values, which is the point: a shape assertion is not a causality check. A test suite that has only ever been run against correct code demonstrates nothing.

The same principle is applied to the split logic, where a known-good split is corrupted one property at a time and the validator is required to catch each corruption specifically.

7. Work in progress — Stage 3

Frozen VideoMAEv2-Base is used as the RGB feature extractor, producing one feature vector per frame. Because the backbone consumes short clips rather than single frames, features are computed on a snippet grid and expanded to per-frame resolution by holding the most recent available value. Every part of that scheme is chosen for causality: each clip covers frames ending at the frame it describes rather than centred on it; the grid is anchored at frame 0 rather than at the end of the video, so that a prefix's grid is a prefix of the full grid; and values are held forward rather than interpolated, since interpolation would blend in a value computed from frames that have not yet arrived.

Feature normalisation is deliberately *not* applied at this stage. Statistics fitted over a whole dataset are forbidden, so mean and standard deviation must be fitted on the training split alone, at the stage that trains. Normalising into the cache would embed a leak where no downstream check could observe it.

The available development machine has no CUDA-capable GPU, so extraction is performed on a free cloud GPU. The 9 GB of frame archives have been uploaded and verified byte-for-byte; extraction is sharded and resumable to fit within the provider's session limit. The resulting cache is checksummed per video, with the backbone, preprocessing recipe and every extraction setting recorded alongside it, so that a copy transferred to whatever machine later trains can be proven to be the cache the recorded settings produced.

Outstanding for this stage: the extraction run itself, and the pose stream, which has not been started.

8. Planned work

- **Stage 4 — baselines and the evaluation harness.** B0 is a trivial reference. B1 is the sliding-window classifier proposed in supervision (2 s window, 0.5 s stride), implemented properly rather than as a straw man: it is the comparison the proposed model must beat and it belongs in the ablation table. The full metric set is built here and used unchanged by every later stage.
- **Stage 5 — the proposed model (M1).** A MiniROAD-style causal GRU with loss computed only at the final position of each training clip, so that training matches streaming inference. Two streams (pose and RGB) with learned late fusion kept swappable for ablation; temporal label smoothing as a ramp at gesture boundaries, since onsets are fuzzy over roughly 0.3 s; class-balanced or focal loss; and clip sampling that guarantees a fixed fraction of training clips contain a gesture onset, without which the gradient signal on onsets vanishes.
- **Stage 6 — the online decision layer.** Frequently omitted in published work, and where much of the practical accuracy lies. Causal smoothing, dual-threshold hysteresis, k-of-n persistence before firing, a refractory period so that one gesture produces one alert, and temperature calibration on validation so that thresholds are meaningful. Every parameter exposed in configuration.
- **Stage 7 — metrics.** Reported together every time: per-frame mAP; point-level mAP under the ODAS protocol swept over onset tolerance; event-level precision, recall and F1; false alarms per hour; median and 95th-percentile onset-to-alarm latency; end-to-end throughput and peak memory. The headline figure plots latency against false alarms per hour, one curve per persistence setting.
- **Stage 8 — analysis.** Confusion matrix and per-class breakdown, a failure-mode montage rendering the worst false positives as actual frames, and an ablation table covering B0/B1/M1, pose-only against RGB-only against fused, with and without label smoothing, and with and without the decision layer — every number on a subject-disjoint split with a seed-variance estimate.

Phase 2 then substitutes our own recorded SOS data behind the dataset adapter. The adapter boundary has been maintained throughout precisely so that this requires one new adapter and one configuration file, with no change to training, model or evaluation code.

9. Risks and open items

- **Training hardware is not settled.** Feature extraction is handled by free cloud GPU and the cache is portable by design, but the hardware for Stages 5 to 8 remains the principal scheduling unknown.
- **Class composition differs from expectation.** In IPN Hand the non-gesture class is the largest by instance count but only the third largest by frame count; two 'pointing' classes are longer resting or transition states. Those three classes together account for roughly 82% of all frames. This affects class balancing in Stage 5 and the false-alarm denominator in Stage 7, and our own SOS recordings will not share this shape — so anything tuned tightly to it will not transfer.
- **The pose stream is not started.** The preferred whole-body pose estimator depends on a toolchain that is awkward to install; a documented CPU fallback exists.

Appendix A — commit history

Commit	Date	Subject
890cfe7	2026-09-24	Stage 3: Kaggle upload staging fixes
f5b13d5	2026-09-20	Stage 3: stage the Kaggle upload
6c4f7bf	2026-09-20	Stage 3 (partial): causal feature extraction, cache, Kaggle path
a9a58e3	2026-09-20	Stage 2: frozen subject-disjoint splits and LOSGO folds
a62f8ef	2026-09-16	Stage 1: IPN Hand adapter and published-figure verification
a08a17c	2026-09-15	Stage 0: canonical data format, config system, run provenance
b165a61	2026-09-15	Scaffold repo: project contract, README, gitignore

Appendix B — reproduction

The repository README carries the commands for every completed stage. The sequence that reproduces the work described above is:

```
python scripts/prepare_ipn_hand.py --zips <download dir> --set dataset=ipn_hand
python scripts/make_splits.py --set dataset=ipn_hand split=ipn_official
pytest # full suite
pytest -m causality # Hard Rule 1
pytest -m splits # subject disjointness
```

Every reported quantity in this document was generated from the repository at the commit named on the first page, rather than transcribed by hand.